INCH-POUND

MIL-D- 87994 (84)

9 Feb 90

MILITARY SPECIFICATION

DISTRIBUTOR, BITUMINOUS MATERIAL, TANK-TYPE, TRUCK MOUNTED

This specification is approved for use within WR-ALC Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.2 Scope. This purchase description covers a tank-type distributor of bituminous material that is mounted on a commercial truck chassis.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Military

KKK-T-2108H	Trucks and Truck Tractors: Commercial, Diesel or Gasoline Engine Drive, 24,000 to 32,000 Pounds GVW, 4 X 2

STANDARDS

Federal

FED-STD-595	Colors

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRG, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3895

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Military

MIL-STD-130	Identification Marking of US Military Property
MIL-STD-129	Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

SOCIETY OF AUTOMOTIVE ENGINEERS

J336	Sound Level for Truck Cab Interior
J534	Lubrication Fittings
J551	Performance Levels and Methods of Measurement of Electromagnetic Radiation from Vehicles and Devices
J638	Tire Chain Clearance - Trucks, Buses
J925	Minimum Service Access Dimensions for Off-Road Machines

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York NY 10001.)

US DEPARTMENT OF TRANSPORTATION (DOT)

Federal Highway Administration

The Motor Carrier Safety Regulation

(Applications for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington DC 20402.)

AMERICAN WELDING SOCIETY (AWS)

Standard Qualification Procedure

Welding Handbook

(Applications for copies should be addressed to the American Welding Society, 345 East 47th St, New York NY 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Welder Qualifications

(Applications for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th St, New York NY 10007.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 First production sample. The contractor shall furnish a complete distributor for first production inspection in accordance with Section 4.

3.2 Description. The item shall consist of a truck mounted bituminous distributor with a hydrostatically driven bituminous pump. The distributor shall be the current commercial production model of the standard product of the manufacturer. The distributor shall be complete with all components supplied as standard equipment to commercial customers together with such attachments as may be necessary for proper function in the environments specified herein. Distributor operation, including all functions and adjustments normally made during bitumen application, shall be able to be accomplished by not more than two operators. The distributor shall be capable of the following functions:

a. Pump bitumen from an external source into the bitumen tank.
b. Pump bitumen from an external source to another external source without bitumen entering the tank or the spray bar.
c. Pump bitumen from the bitumen tank back to the bitumen tank without the bitumen entering the spray bar.
d. Pump bitumen from the bitumen tank through the spray bar back to the bitumen tank.
e. Pump bitumen out of the bitumen tank to an external source without the bitumen entering the spray bar.
f. Pump bitumen out of the bitumen tank through the spray bar and out of the nozzles.
g. Pump bitumen out of the bitumen tank to the hand spray attachment.
h. Withdraw bitumen from the spray bar and all piping and return it to the bitumen tank.
i. Pump flushing oil through all piping and the spray bar without the flushing oil entering the bitumen tank.
j. Withdraw bitumen from the hand spray attachment or blow out hand spray attachment.

3.3 Material. The components, pieces and parts incorporated in the distributor may be newly fabricated from recovered materials to the maximum extent practical, provided the distributors produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the distributor. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials.

3.4 Environmental requirements. The distributor shall perform as specified in any ambient temperature from +32°F to +115°F. The truck shall be capable of starting down to ()°F and operation down to -20°F. Neither the truck nor the distributor hardware shall be damaged by storage between -40°F and 125°F. The unit shall withstand wind and wind blown precipitation. Bitumen shall not be contaminated by precipitation or dust and operation shall not be degraded due to extended outside storage in these environments.

3-5 Safety. All exposed parts which are subject to high operating temperatures or which are energized electrically shall be insulated, fully enclosed, or guarded. Protective devices shall not impair the operational functions. These devices shall be removable in order to facilitate maintenance or service operations.

3-5.1 Steps, grabrails, and platforms. Steps, grabrails, platforms, etc shall be provided to ensure safe and convenient access to all control, maintenance, and inspection points.

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3-5.2 Noise level. The distributor truck shall meet the following requirements when tested in accordance with 4.4.5:

a. Cab interior sound level not greater than 84 dB(A).
b. Exterior operator station sound level not greater than 94 dB(A).
c. If either of the above requirements is not met, the distributor truck shall be inspected to ensure all standard and optional sound damping equipment which would substantially reduce the sound level at the point(s) which fail to meet the requirements above are present and properly installed. Engine enclosure, engine enclosure insulation, cab insulation, and engine exhaust system shall be inspected to determine if overall sound level reductions can be made. If sound level remains in excess of the above limits, a permanent warning label shall be attached in a conspicuous location. The label shall state that hearing protection is required when machine is in use. This paragraph is not intended to require major system or configurational changes to the unit; but, to ensure that reasonable steps have been taken to minimize the noise hazard. The installation of additional insulation, baffles, better mufflers, and sheet metal stiffeners and isolators, where effective in reducing noise level, shall be considered required measures to minimize noise hazard in excess of stated limits.

3.5.3 Sharp edges. All nonfunctional edges shall be rounded or broken so as to not present a safety hazard.

3.6 Maintainability. All maintenance described in this specification shall be capable of being performed by one mechanic using common hand tools. Design and construction of the distributor truck shall permit ready accessibility to all parts for maintenance and repair in the field. Access shall be in accordance with SAE J925 minimum access dimensions. Replacement and adjustment of component parts and accessories shall be accomplished with minimum drainage of fluids and disturbance to other parts.

3.6.1 Filling, draining, and checking provisions. All fluid systems shall possess a convenient means of filling, draining, and checking the level of the system. All drains shall conduct fluid directly to the ground in a stream without spilling onto other components. Fluid level indicators or fill ports shall identify the type of fluid, state whether the level should be checked cold, hot, or with engine running, and identify appropriate add and full levels.

3.6.2 Lubrication points. Lubrication points shall be provided for all surfaces requiring lubrication. Pressure relief fittings shall be supplied where the use of pressure lubrication equipment may damage seals. Lubrication fittings shall be in accordance with SAE J534 and be located so as to be clearly visible and easily serviced with a grease gun with a 10 inch flexible hose.

3.6.3 Maintenance operations list. Any four or more operations shall be able to be performed by one person in the sum of times allotted for each operation.

a. Remove, replace, and adjust all engine driven belts - 1 hour.
b. Remove and replace voltage or current regulator - 15 min.
c. Remove and replace alternator and adjust alternator belts - 30 min.
d. Remove and replace all filter elements, screens, and strainers on the hydraulic system - 30 min.
e. Drain cooling system, remove and replace engine cooling hoses, and refill cooling system - 1 hour.
f. Drain engine lubricating oil, remove and replace oil filter elements(s), and refill crankcase - 30 min.
g. Remove and replace fuel filter elements - 30 min.
h. Remove and replace transmission fluid filter elements and strainer - 45 min.
i. Drain and refill transmission and torque converter - 20 min.
k. Remove cover, disconnect battery leads, remove and replace batteries, reconnect battery leads, and replace cover - 20 min.

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3-7 Truck. The distributor shall be mounted on a commercial truck chassis. The truck shall conform to KKK-T-2108H, Type I, Class A, and shall have an automatic transmission. Ordering data for this specification will be provided in the solicitation or from the contracting officer upon request. If the requirements stated in this document differ from those in KKK-T-2108H, the requirements in this document take precedence.

3-8 Bitumen tank. The tank shall be an elliptical cylinder constructed of at least Number 10 gage (.1345 inch nominal thickness) sheet steel. The heads shall be constructed of at least Number 8 gage (.1644 inch nominal thickness) sheet steel reinforced, concave or convex, and all welded seams shall be bitumen tight. Surge plates, of the same gage steel as the tank, shall be provided and welded in place. The distances between surge plates or between surge plates and tank ends shall be not more than 60 inches. Surge plates shall be the full size of the tank cross section, and shall be dished or reinforced to withstand surge pressures. Openings in the surge plates shall be large enough to provide free venting at the top, free flow of material at the bottom, and access to all sections of the tank. Access holes in the surge plates shall be staggered. The tank shall be provided with a manhole large enough to permit free access to the tank. The manhole shall be equipped with a basket type, removable steel strainer. A quick-clamp-type cover, which is bitumen tight and can be opened with one hand, shall be provided. The cover shall be domed to prevent accumulation of water. The cylinder and heads of the tank shall be covered with at least 2-inch thickness of the manufacturer's standard commercial insulating material. The insulating material shall be installed in a manner as to prevent packing or shifting during use or travel. The insulation shall be protected by a water tight sheath constructed of at least Number 20 gage (.0359 inch nominal thickness) sheet steel. The tank shall have a capacity of at least 800 gallons, plus four percent for expansion.

3.8.1 Overflow. A 3-inch overflow shall be provided for the tank. The overflow shall be taken off above the top of the tank near its centerline and shall conduct the fluid to a point beneath the tank. The manhole may serve as the overflow dome. The bottom of the overflow shall be provided with and extension to prevent any discharge coming in contact with any part of the chassis including the drive train.

3.8.2 Drain. Means shall be provided to completely drain the bitumen tank at such a location that the discharge will clear the distributor.

3.9 Heating system. The heating system shall consist of flues, burners, fuel system, blower, and a portable burner.

3.9.1 Heating flues. The tank shall be equipped with either U-type or continuous heating flues with a minimum outside diameter of 8 inches at the burner end. When U-type flues are provided, there shall be not less than two sets extending the entire length of the tank. Flues shall be of seamless steel tubing, welded steel tubing, or spiral butt-welded pipe so arranged as to effectively heat the bitumen. The flues shall be supported in the tank in such a fashion that vibration will be minimized and yet permit expansion and contraction.

3.9.2 Burners. Heat shall be supplied by two low pressure, air atomizing type burners using commercial grade diesel fuel. Burners shall be accessible for adjustment and cleaning. The tank shall be equipped with at least one of the following: Shield(s), flue liners, or combustion chambers. No flame shall come in direct contact with the flues or tank. Heat supplied by the burners shall be sufficient to raise the temperature of a full tank's contents at least 1°F per minute. Fuel consumption for each burner shall be adjustable to include all flow rates from shutoff to maximum required fuel consumption. Any equipment required for igniting the burners shall be supplied. If this equipment is handheld, storage provisions shall be provided on the truck near the burners.

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3.9.3 Fuel tank. The fuel tank shall be manufacturer's current standard production tank with a minimum capacity of 25 gallons. The tank shall be marked "FILL WITH DIESEL FUEL ONLY." The tank shall be located or positioned to minimize fire hazard caused by spillage, overflow, or drainage. Design of the fuel tank shall include an easily accessible drainage valve. The tank shall be vented to permit easy filling.

3.9.4 Fuel pump. A pump shall be provided to supply fuel to both burners at constant pressure. The pump shall be connected to the power source so that pumping will be accomplished only when the burner blower is operating.

3.9.5 Fuel line. The fuel line shall be of seamless metal tubing or oil resistant hose, and shall be equipped with a screen strainer and a bypass or a combination bypass strainer, a pressure regulator, a pressure gage, and a valve for regulating the volume of fuel supplied. The fuel lines shall be mounted so as to prevent damage from vibration.

3.9.6 Blower. A blower shall be provided to supply enough air for proper combustion when both burners are operating at full capacity. The blower shall be of the positive-pressure, rotary type. Blower operation shall be independent of the bitumen pump.

3.9.7 Portable burner. A portable generating-type burner designed to use commercial grade diesel fuel shall be furnished. The burner shall be used for heating the asphalt pump or the tank car connections. The burner at maximum firing rate shall burn not less than .5 gallon of fuel per hour and shall have a fuel-tank capacity of not less than 4 gallons. The tank shall be marked "FILL WITH DIESEL FUEL ONLY." Provision shall be made for carrying the burner on the distributor.

3.10 Pumping system. The pumping system shall include a positive displacement rotary type bituminous pump having a capacity of not less than 375 gallons per minute. Provisions shall be made for heating the pump with the portable burner. The pumping system shall provide bitumen to the spray bar to produce the application rates specified in 3.15.

3.11 Hydrostatic transmission. The bituminous pump shall be driven by a hydrostatic transmission. The hydrostatic transmission shall have a ratio range which allows the application rate to be varied as previously specified. The hydrostatic transmission shall be designed to tie the truck speed to the distributor pump output in such a manner that the pump output is proportional to truck speed.

3.12 Spray bar and piping. The spray bar shall be a turn-up, full circulating type and shall consist of a center section and two fold-up end sections, with provisions to add positive locking extension sections to each end of the spray bar to make a total spray bar length of 24 feet. An extension section shall be not greater than 2 feet long. The spray bar shall be designed to meet the distribution requirements of paragraph 3.15. Provisions shall be made to fold the extended portion of the spray bar for travel. The spray bar shall lock in both the folded and extended positions. The spray bar shall be adjustable for road crown. The design of the spray bar shall be such that when the bar strikes an obstruction while the distributor is moving forward, it will bend, thus lessening chance for damage to the spray bar or components. Nozzles shall be V-slot, dripless, nonclogging, on 4 inch centers and shall be provided with individual cut-off valves. All nozzle valves shall be linked together so that they can be opened or closed simultaneously by a single lever operation. The spray bar shall be laterally and vertically adjustable by means of mechanical linkages or hydraulic controls. The spray bar shall be capable of being shifted at least 8 inches either side of center. Bar shall provide for vertical adjustment from the travel position downward. All valves subject to pumping pressure shall be the leakproof type.

3.13 Controls systems and indicators. All necessary controls and indicators for efficient and controlled operation shall be installed. Controls and indicators shall be visible to, and within reach of, an operator in the normal operating position. All controls shall be clearly identified with permanently affixed identification. Material used for identification marking shall be corrosion resistant. Control systems shall be provided to permit all of the operations listed in 3.2.

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3.13.1 Cab controls. In addition to the cab controls furnished by the chassis manufacturer, there shall be a control console containing controls for all distributing functions. These shall include, but are not limited to the following:

a. Start and stop spraying.
b. Raise/lower spray bar between travel and operating positions.
c. Sideshift spray bar.
d. Raise/lower ends.
e. Application rate control.
f. Travel indicator wheel raise/lower control.
g. PTO engage/disengage (may be located in the dash or with the other distributor controls).

3.13.2 Cab indicators. In addition to the gages and indicators furnished by the chassis manufacturer, the following indicators shall be installed in the cab and be visible to the operator while seated in the normal operating position:

a. Rate of travel indicator with ft/min scale. Operating range of 0 to 1500 ft/min minimum with increments of not more than 20 ft/min. Accuracy shall be 5% of indicated speed for any speed over 100 ft/min.
b. Cumulative distance indicator to 999,999 feet.
c. Resetable distance indicator to 9,999 feet. Accuracy shall be five percent of any indicated distance greater than 50 feet.

The preceding indicators shall be driven from a ground contact wheel which can be raised and lowered by a control in the cab. The wheel shall lock up for transport.

d. Bitumen application rate in gal/min. Meter shall be driven by a pump tachometer.
e. PTO tachometer scaled in RPM.

3-13.3 Operator's station controls. Controls for all distributing functions shall be operable from the operator's station. These shall include, but are not limited to the following:

a. Start and stop spraying. Distributor shall go to circulate mode when spraying stops.
b. Raise/lower spray bar between travel and operating positions.
c. Sideshift spray bar. Bar shall be able to be shifted at least 8 inches either side of center.
d. Fold up/unfold ends.
e. Start/stop air blowout of the plumbing, spray bar, and hand-held sprayer as applicable.
f. Start/stop the withdrawal of bitumen from the plumbing, spray bar, and hand-held sprayer as applicable.
g. Start/stop the flushing oil in the plumbing and spray bar and eliminating the flushing oil from the system.

3-13.4 Operator's station indicators. At least the following indicators shall be easily readable from the operator's station:

a. Bitumen volume gage. Gage shall have a range from full to empty and be calibrated in increments not greater than 25 gallons. Accuracy shall be within 25 gallons for any tank level.
b. Bitumen temperature gage with a Fahrenheit scale. Range shall be 50° to 500°F minimum.

3.13.5 Other controls and indicators. The controls for redirecting bitumen flow to switch between such functions as onloading and offloading bitumen, and for other functions not directly related to spraying bitumen under normal conditions, shall be operable either from the ground or from the operator's station. The spray bar shall be adjustable to compensate for changes in road camber. The spray width shall be adjustable to widths of 4 and 8 feet (+/-4 inches), and in increments of not greater than 1 foot between 8 feet and 24 feet. None of these operations shall require the use of tools or require more than five minutes to achieve desired settings.

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3.14 Distribution. The width of distribution shall be 4 feet, 8 feet, and from 8 feet through the full width of the spray bar in increments of 1 foot. The rates of application shall be from 0.1 gallon through 1.0 gallons per square yard in increments of not more than 0.1 gallon per square yard. The truck-mounted distributor shall apply bitumen through the spray bar to the surface being sprayed at all combinations of specified widths and rates. The accuracy of distribution over a given area shall be within plus or minus 5 percent. The application at all rates shall be accomplished without streaking or puddling. The pump shall be capable of maintaining a constant desired flow rate over the entire useful range of flowrates in both the "recirculate in spray bar" and "spray" modes, sufficient to pass the tests in 4.4.6.

3.15 Distribution chart. Two distribution charts shall be provided with each distributor. The charts shall provide the following for each combination of specified spray widths and spray rates.

a. Metering device reading or setting.
b. Travel speed in feet per minute (fpm).

The charts shall be coated with a clear plastic material for protection.

3.16 Accessories. Two filling hoses, a tank car valve and couplings, a complete hand spray attachment, and fittings shall be furnished with each distributor. Provisions shall be made to carry the accessories on the distributor when they are not in use. Accessories shall remain in their carrying places when the distributor is transported or operated as specified herein.

3.16.1 Filling hose. The filling hoses shall be the manufacturer's standard commercial product; however, they shall be at least 15 foot lengths of 3 inch flexible, high temperature type. The hoses shall be fitted on both ends with hose-end components for 3-inch quick-disconnect coupling.

3.16.2 Tank-car valve and couplings. A tank-car valve shall be furnished in the loading line and shall be vented. The valve shall be either a quick-acting gate valve or a lever-operated plug valve. The loading line and transfer line both shall be fitted with faucet end and chain secured dust cap components for a 3 inch, quick-disconnect coupling.

3.16.3 Hand spray attachment. The hand spray attachment shall consist of a spray tube with nozzle, handle and control cock, cut-off valve, and a half coupling for connection to the hose. The hose shall be not less than 1 inch in diameter. The hose shall be 25 feet long, asbestos lagged, and furnished with quick-disconnect on each end. The spray line on the distributor shall be equipped with a faucet end half coupling for connection to the hose and a chain secured dust cap.

3.16.4 Fittings. All components of the 3 inch, quick-disconnect coupling specified herein shall be manufacturer's current standard production. The following additional components shall be furnished.

a. Two double adapters for a 3 inch, quick-disconnect coupler.
b. One faucet with chain secured dust cap for a 3 inch quick-disconnect coupler.

3.17 Platform, catwalks, and steps. Platforms, catwalks, and steps shall be made of nonskid expanded or perforated metal surface. These items shall not restrict access to the truck fuel tank or bitumen tank.

3.17.1 Operator's platform. A platform, having access to all distributor controls, shall be provided for the operator.

3.17.2 Catwalks. Full length catwalks shall be provided along both sides of the bitumen tank. The truck chassis fuel tanks shall in no way be used to gain access to the catwalks.

3.17.3 Steps. Steps shall be provided on both sides of the distributor for access to the operator's platform. When the catwalks and operator's platform are at different elevations, steps shall be provided between platform and catwalks. Steps shall be not less than 5.5 inches wide and shall be flanged. A ladder or rungs shall be provided on the tank for access to the manhole on top of the tank.

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3.18 Grabrails or guardrails. Steel pipe grabrails shall be mounted on both sides of the bitumen tank and shall extend the full length of the tank. The grabrails shall be of a height to be accessible to the operator when he is standing on a catwalk or going from the operator's platform to a catwalk. Guardrails shall be installed around the perimeter of the operator's platform. Guardrails need not be installed at perimeter locations where a part of the distributor restricts access by the operator. Guardrails shall have openings at step locations to provide access to and from the platform and catwalks, and to and from the ground platform. Detachable chains shall be provided at openings.

3.19 Fenders. Design of the distributor shall include fenders for the rear wheels on each side truck. These fenders shall be constructed of sheet steel not less than 0.1196 inch nominal thickness. The tire to the fender clearance shall be in accordance with SAE J683. The fenders shall cover the width of the tires. The top of the fenders may serve as catwalks.

3.20 Signal system. A mechanical signal bell or electric signal system, operable from both the truck's cab and the operator's platform, shall be furnished. Should the operator's platform be located at the front of the distributor, a signal system is not required.

3.21 Electromagnetic interference. The electromagnetic interference emission characteristics of the distributor shall conform to the electromagnetic interference requirements and test limits of SAE J551a.

3.22 Lubrication. All bearings (except sealed-for-life type), mating gears and sliding parts shall be provided with means for lubrication. Lubrication fittings shall be in accordance with SAE J534. Reservoirs shall be provided with means for inspection of oil levels. All oil holes, grease fittings, and filter caps shall be easily accessible. Any parts which may be damaged by the use of pressure lubrication equipment shall be provided with pressure relief devices.

3.23 Lifting attachments. Permanently affixed attachments that enable the unmounted distributor to be lifted to its normal travel position shall be provided. Each attachment shall withstand, without damage to any part of the unmounted distributor, 2.5 times the resulting vectorial force imposed when the unmounted distributor is suspended by the lifting attachments. Attachments for multiple slings shall be located so that: (a) Slings shall converge at not more than 19 feet above the lowest extremity of the unmounted distributor; and (b) not less than 1-inch clearance will be maintained between the slings and the unmounted distributor. Spreader bars are permitted. The inside diameter of the attachment eye shall be not less than 3 inches. The lifting attachments for the unmounted distributor shall be located on top of the tank at both ends of the tank. A caution plate shall be attached to the distributor to indicate that the unmounted distributor lifting attachments are not for use in lifting the complete truck mounted distributor. The complete truck mounted distributor shall be lifted by attaching slings to the shackles provided on the front of the chassis and to the lifting attachments on the axle assembly without damaging any part of the distributor.

3.24 Tiedown attachments. Permanent affixed attachments that permit fastening the mounted distributor to the floor or deck of a transportation medium shall be provided. Each attachment shall withstand, without damage to the mounted distributor, its proportion of the following static loads: 4.5-G forward and rearward, 2.25-G upward, 1.5-G sideward, where G equals weight of the mounted distributor. Attachments shall be rounded and beveled to prevent cutting of the tiedown material. The inside diameter of the attachment eye shall be not less than 3 inches.

3.25 Data plates. The following plates shall be installed in a conspicuous location on the end item.

3.25.1 Identification plate. A corrosion resistant metal name plate shall be permanently and legibly marked, in accordance with MIL-STD-130, with the following information:

a. Nomenclature.
b. Manufacturer's model number.

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c. Manufacturer's serial number.
d. Special characteristics (HP, RPM,. etc).
e. National stock number.
f. Manufacturer's name.
g. Date of manufacture.
h. Warranty expiration date.
i. Curb weight and CG location.

3.25.2 Lubrication plates. A corrosion resistant metal lubrication plate(s) shall illustrate the lubrication points on the vehicle and shall identify the type of lubricant and service interval for each point.

3.25.3 Warning plates. Warning plates shall identify potentially hazardous conditions, locations, and actions. Plates shall be sized and located so as to be conspicuous to the person at risk.

3.25.4 Instruction plates. All controls shall be labeled to identify the location and function of each control position. Instructions cast or stamped on or beside the controls are acceptable.

3.26 Treatment and painting. The rear of the bitumen tank, stack, pump, pump housing, and all piping subject to heating by portable burner during normal start up shall be painted with a silicone base, heat-resistant, flat black paint. All other portions of the distributor normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard commercial practice, except that the finish shall be in accordance with FED-STD-595, Chip No. 24052, forest green. All trim such as door handles, light trim rings, etc shall be coated to match the cab color. The tank skin shall be painted to match the cab if temperature conditions permit, or shall be coated with the flat black heat resistant paint, if necessary.

3.27 Stenciling. The gross vehicle weight of the truck-mounted distributor shall be applied on each side of the distributor. The prescribed tire pressure shall be stenciled near each tire. All lifting and tiedown attachments shall be marked "Lift Here" and "Tiedown Here" as applicable. Registration numbers shall be marked on the truck-mounted distributor. Size of letters and numerals to be used are as follows:

US AIR FORCE 1.5 INCHES
REGISTRATION NUMBER 1.5 INCHES
REG NUMBER SPACING EXAMPLE 77D1977

3.28 Tools and lubricating equipment. Tools normally furnished with the distributor by the manufacturer shall be provided together with the manufacturer's regular tool storage space or box.

3.29 Electrical system. The electrical system for the truck mounted distributor shall consist of the truck manufacturer's commercial electrical system and the additional components and wiring necessary to meet all requirements of the DOT Motor Carrier Safety Regulations. All lights shall be mounted in protected locations, and shall remain visible under all conditions of operation. Identification, sidemarker, and clearance lights shall be of the armored or flush-mounted type.

3.30 Workmanship. Workmanship of the distributor, allied equipment, and accessories shall be of the quality prevailing among manufacturers normally producing equipment of the type specified herein.

3.30.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock washers shall be provided in accordance with good commercial practice. All bolts, ruts, and screws shall be tight.

3.30.2 Welding. The surface of part to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds for strength will not be permitted. Weld penetration shall be such as to provide transfer of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Welding shall be in accordance

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with a nationally recognized welding code. Additionally, before assigning any welder to manual welding work covered by this purchase description, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

a. Standard Qualification Procedure of the American Welding Society.
b. Welding Qualifications of the ASME.

3.30.3 Castings. Castings shall be sound and free from patching, misplace coring, warping, or other defects which might render the casting unsound for use.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First production inspection.
b. Quality conformance inspection.

4.2.1 First production inspection. First production inspection shall be applied to the sample submitted in accordance with 3.1. Failure of the first production sample to pass the examination or one or more first production tests shall be cause for rejection. First production inspection shall consist of the examination in 4.4.1 and all tests in 4.4.2 – 4.4.8

4.2.2 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Failure of the machine to pass the examination or one or more tests shall be cause for rejection. Quality conformance inspection shall consist of the following:

a. Examination (see 4.4.1).
b. Test (see 4.4.2 and 4.4.8).
c. Inspection of preservation, packaging, packing and marking for shipment storage (see 4.5).

4.3 Test conditions.

4.3.1 Apparatus. The apparatus used in conjunction with the testing specified herein shall be of laboratory precision and shall be calibrated at proper intervals to ensure accuracy.

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4.3.2 Preparation for tests. The vehicle, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.3 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, any of which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.
b. Spillage or leakage of any fluid, including fuel, coolant, lubrication or hydraulic fluid, under any operation condition except as allowed herein.
c. Structural failure, evidence of impending failure, or permanent deformation of any component or joint.
d. Evidence of excessive wear.
e. Interference between vehicle components or between the vehicle and the ground (including all required obstacles).
f. Misalignment of any component.
g. Evidence of undesirable roadability characteristics, including instability in handling while cornering, braking, and traversing rough terrain as required.
h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.
i. Overheating of the engine, transmission, or any other vehicle component.
j. Failure of any control to establish and maintain desired settings.
k. Failure of any indicators to provide readable, accurate, or meaningfull measurement.

4.3.4 Notification of testing. The Government shall be, in accordance with contract provisions, notified of all testing to be performed in fulfillment of the requirements of this specification so that arrangements may be made to witness the testing. Addresses and specific requirements of notifications shall be provided in the solicitation, contract, or by the Government's contracting officer upon request.

4.4 Tests.

4.4.1 Examination. Each distributor shall be examined for compliance with the requirements specified in Section 3. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failures to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examination and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects shall constitute cause for rejection. Items checked in Column 1 of Table I shall be inspected during first production inspection. Items in Column 2 shall be inspected during acceptance inspection.

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TABLE I

F/P	ACC	DESCRIPTION		PARA NO.
X	X	101	Distributor configuration not as specified.	3.2
X	X	102	Material not as specified.	3.3
X	X	103	Safety not as specified	3.5, 3.5.1, 3.5.3,
X		104	Access for maintenance not as spec.	3.6
X		105	Fluid filling, draining, and checking provisions not as specified. engine oil ( ) transmission oil ( ) coolant sys. ( ) truck fuel tank ( ) pump hyd. oil ( ) burner fuel tank ( ) bitumen tank ( ) air tank drain ( ) portable burner fuel tank ( )	3.6.1
X	X	106	Lubrication provisions not as spec.	3.6.2
		107	Truck not as specified.	KKK-T-2108H
			Type and Class not as specified.	1.2 OF 2108
X			General design not as specified.	3.2 of 2108
X	X		Chassis components not as specified.	3.4 of 2108
X	X		Servicing not as specified.	3.6 of 2018
X		108	Tank not as specified.	3.8
X		109	Overflow not as specified.	3.8.1
X		110	Heating system configuration not as specified.	3.9 - 3.9.7
X		111	Pumping system configuration not as specified.	3.10
X		112	Hydrostatic transmission not as specified.	3.11
X	X	113	Spray bar and piping not as specified.	3.12
X	X	114	Controls and indicators not as specified.	3.13- 3.13.5
X	X	115	Distribution chart not as specified.	3.15
X	X	116	Accessories not as specified.	3.16 - 3.16.4
X	X	117	Platform, catwalk, and steps not as specified.	3.17 - 3.17.3
X	X	118	Grabrails and guards not as specified.	3.18
X	X	119	Fenders not as specified.	3.19
X	X	120	Signal system not as specified.	3.20
X	X	121	Lifting attachments not as specified.	3.23
X	X	122	Tiedown attachments not as specified.	3.24
X	X	123	Dataplates not as specified.	3.25 - 3.25.4
X	X	124	Painting not as specified.	3.26
X	X	125	Stenciling not as specified.	3.27
X	X	126	Tool and lubricating equipment not as specified.	3.28
X	X	127	Electrical system not as specified.	3.29
X	X	128	Workmanship not as specified.	3.30 - 3.30.3

4.4.2 Performance test. Fill distributor with high-viscosity oil and operate the distributor for a run-in period of 60 minutes. Turn on all lights. Check all cab controls to ensure proper function. Check the distributor for leaks. Operate of all distributor controls and check for proper function. Make necessary adjustments prior to shipment. Any leak or malfunction of any component that is not corrected by adjustment shall constitute failure. Drive truck approximately one mile without using brakes. Coast to a stop and inspect the wheels for heating (indication of brakes dragging). Accelerate hard to 25 mph and

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brake hard to stop. Repeat for ten cycles with no pauses. There shall be sufficient air pressure to release the brakes after the tenth run.

4.4.3 Weight distribution. Weigh each axle of the fully loaded distributor. The weight on each axle shall not exceed 95 percent of the axle weight rating.

4.4.4 Heating test. The unit shall be filled with a high viscosity oil. The burners shall be lit and the oil circulated. Allow the temperature of the oil to rise at least ten degrees. Start stopwatch and record the time required to raise the oil temperature 20 degrees. Time greater than 20 minutes shall be cause for rejection.

4.4.5 Noise level testing.

a. Compliance with the requirements of 3.5.2.a shall be verified by tests in accordance with SAE J336.
b. Compliance with the requirements of 3.5.2.b shall be verified by tests conducted as follows:

(1) Instruments and site shall be same as above.
(2) Microphone shall be located at the position of the head of the operator who would be using the external controls. The vehicle shall discharge oil through the bar and both extensions at the highest recommended rate of discharge for this width.
(3) If either a. or b. above fail to meet requirements, 3.5.2.c. shall take effect.

4.4.6 Test strip. This test is intended to ensure the ability of the pump to maintain a constant flow rate when switching between circulate and spray modes. Set the spray nozzles for a 4' spray. Determine the minimum recommended spray rate for the width/nozzle combination. With distributor in the "circulate in spray bar mode", set the pump rate. Switch to spray mode and record any change in pump flow rate. Repeat with maximum recommended flow rate. Repeat for 8' width and full width spraying, for a total of 6 runs. A change in flow rates of more than five percent in any reading shall be cause for rejection.

#	width	recommended flow rates	actual recirc. flow rate *1	actual spray flow rate *2	% change (1-2)/1*100
1	4 feet	min			
2	4 feet	max			
3	8 feet	min			
4	8 feet	max			
5	full	min			
6	full	max			

4.4.7 Maintenance testing. After having performed the inspection according to 4.4.1, an authorized government representative shall select between four and all of the tasks listed in 3.6.3. These tasks shall be performed sequentially by one person (unless otherwise indicated in 3.6.3) in the sum of the time allotted for each task. All tools, parts, and supplies shall be on hand at the start of the test. The list of tasks performed and the total time taken to perform the tasks shall be recorded. Power tools shall not be used.

4.4.8 Final inspection. The unit shall be examined for signs of any conditions for failure listed in 4.3.3.

4.5 Inspection of preservation, packaging, packing and marking for shipment and storage. Preservation, packaging, packing and marking for shipment and storage of the item shall be inspected to verify conformance with the terms of the contract or order. A transportation data plate in accordance with MIL-STD-129 shall be installed.

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5. PACKAGING.

5.1 Preservation, packaging, packing and marking. Preservation, packaging, packing, and marking shall be in accordance with the provisions of the contract. A transportation data plate (shipping board) shall be furnished in accordance with MIL-STD-129.

6. NOTES.

(This section contains information of a general or explanatory nature that may be helpful, but it is not mandatory.)

6.1 Intended use. The distributors are primarily intended for use in accurately applying bitumen during road and airfield construction. The distributors can also be used for transporting or transferring bitumens or for application of bitumen for dust alleviation.

6.2 Ordering data. This specification requires no ordering data. KKK-T-2108 ordering data applies to the truck chassis on which the distributor is mounted and must be obtained from the contract, or from the contracting officer upon request.

6.3 First Article. When first article inspection is required, the contracting officer should provide specific guidance to offerors, whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first article. Invitations for bid should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Subject term (key word) listing.

Bitumen tank
Blower
Burner
Electromagnetic interference
Hydrostatic transmission
Spray bar

Custodians:
Air Force - 99

Preparing activity:
Air Force - 84

Project No: 3895-F271

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-87994(84)	Trailer, Rapid Runway Repair

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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